The idea of the rational numbers

Every fraction, positive or negative — the set in which all four operations work.

LESSON 37–38 | 2 × 40 MIN | MATEMATIKA 7, §15 | S8 7.1

LESSON CLOCK

15' | 25' | 25' | 20' | 5'

The definition	15 min
Everything is a fraction	25 min
Fractions and decimals	25 min
On the line	20 min
Homework	5 min

LEARNING OBJECTIVES

- Define a rational number and use the symbol $\mathbb{Q}$.
- Write an integer, a decimal and a mixed number as a fraction.
- Convert between fractions and decimals, including recurring ones.
- Place rational numbers on the coordinate line.

TEXTBOOK

National	pp. 95–100
Cambridge	Stage 8, pp. 66–72
Workbook	Exercise 7.1

01 Explanation

The definition

> A number is rational if it can be written as $\frac{p}{q}$ with $p, q \in \mathbb{Z}$ and $q \neq 0$

The set of them is $\mathbb{Q}$. It contains every integer, every fraction, every terminating decimal and every recurring decimal — positive and negative alike.

OPERATION	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$
addition	✓	✓	✓
subtraction	✗	✓	✓
multiplication	✓	✓	✓
division (not by 0)	✗	✗	✓

$\mathbb{Q}$ IS THE FIRST SET IN WHICH ALL FOUR OPERATIONS WORK

That is the whole reason for the chapter. Grade 8 will find that $\sqrt{2}$ is **not** rational, and the system will have to grow once more.

Everything is a fraction

NUMBER	AS $\frac{p}{q}$
7	$\frac{7}{1}$
-4	$\frac{-4}{1}$
0.75	$\frac{3}{4}$
$2\frac{1}{3}$	$\frac{7}{3}$
-1.6	$-\frac{8}{5}$
0	$\frac{0}{1}$

A fraction may be written in many equivalent ways: $\frac{3}{4} = \frac{6}{8} = \frac{75}{100}$. The **simplest form** has no common factor above 1 in numerator and denominator.

THE DENOMINATOR MAY NEVER BE ZERO

$\frac{5}{0}$ is not a rational number; it is not a number at all. The condition $q \neq 0$ is part of the definition, not an afterthought.

Fractions and decimals

Divide the numerator by the denominator. The result is either terminating or recurring — never anything else.

FRACTION	DECIMAL	KIND
$\frac{3}{4}$	0.75	terminating
$\frac{1}{8}$	0.125	terminating
$\frac{1}{3}$	$0.333\ldots$	recurring
$\frac{2}{11}$	$0.1818\ldots$	recurring
$\frac{5}{6}$	$0.8333\ldots$	recurring

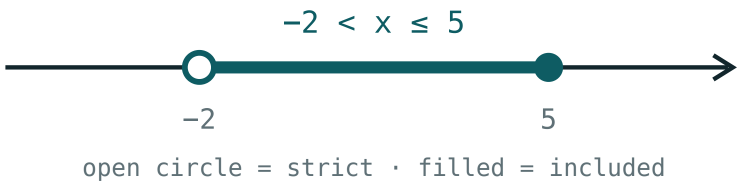
A fraction in simplest form terminates exactly when its denominator has no prime factors other than 2 and 5 — because $10 = 2 \times 5$.

RATIONAL ⟺ TERMINATING OR RECURRING

A decimal that neither stops nor repeats — such as $0.101001000100001\ldots$ — is not rational. Grade 8 will meet a famous one: $\sqrt{2}$.

On the line

Every rational number has a place. To mark $\frac{3}{4}$, divide the unit into 4 parts and take 3; to mark $-\frac{3}{4}$, do the same to the left.



Dividing the unit segment places any fraction exactly.

NUMBER	LIES BETWEEN
$\frac{3}{4}$	*0 and 1*
$-\frac{3}{4}$	*−1 and 0*
$\frac{7}{3}$	*2 and 3*
$-\frac{7}{3}$	*−3 and −2*

BETWEEN ANY TWO RATIONALS THERE IS ANOTHER

Their average. So the rationals leave no gaps a ruler can see — and yet Grade 8 will show that gaps remain.

02 Worked examples

EXAMPLE 1 **Write 7, -1.6 and $2\frac{1}{3}$ as fractions $\frac{p}{q}$.**

1. $7 = \frac{7}{1}$

   Every integer is a fraction.

2. $-1.6 = -\frac{16}{10} = -\frac{8}{5}$

   Simplify.

3. $2\frac{1}{3} = \frac{6+1}{3}$

4. $= \frac{7}{3}$

ANSWER

$\frac{7}{1}, -\frac{8}{5}, \frac{7}{3}$

EXAMPLE 2 **Convert $\frac{3}{8}$ and $\frac{5}{6}$ to decimals, and say which recurs.**

1. $3 \div 8 = 0.375$

   Terminating.

2. $8 = 2^3$ — only twos.

3. $5 \div 6 = 0.8333\ldots$

   Recurring.

4. $6 = 2 \times 3$ — the 3 causes it.

ANSWER

0.375 and $0.8333\ldots$

EXAMPLE 3 **Between which two integers does $-\frac{7}{3}$ lie?**

1. $\frac{7}{3} = 2\frac{1}{3}$

2. So $-\frac{7}{3} = -2\frac{1}{3}$.

3. That is further left than -2.

4. Between -3 and -2.

ANSWER

-3 and -2

03 Interactive model

Ask for a number between 0.33 and 0.34, then between that and 0.34; the class discovers that the process never ends, which is what “dense” means.

Fractions and their simplest form

$$\frac{18}{24}$$

1. $18 = 2 \cdot 3^2$ and $24 = 2^3 \cdot 3$

   Factorise both.

2. The common factor is 6.

   The highest one — anything less leaves work to do.

3. $$\frac{18 \div 6}{24 \div 6} = \frac{3}{4}$$

   Divide both parts by it.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Rational number	Ratsional son	Рациональное число
Fraction	Kasr	Дробь
Numerator	Surat	Числитель
Denominator	Maxraj	Знаменатель
Terminating decimal	Chekli o‘nli kasr	Конечная десятичная дробь
Recurring decimal	Davriy o‘nli kasr	Периодическая дробь
Mixed number	Aralash son	Смешанное число
Equivalent fractions	Teng kasrlar	Равные дроби

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A rational number is:

- any decimal
- $\frac{p}{q}$ with integers, $q \neq 0$
- an integer
- a fraction below 1

Is 7 rational?

- yes
- no
- sometimes
- undefined

$\mathbb{Q}$ is closed under:

- addition only
- all four operations (not ÷0)
- subtraction only
- none

$\frac{3}{8}$ as a decimal:

- 0.38
- 0.375
- 0.83
- 0.125

A fraction terminates when its denominator has only:

- 2 and 3
- 2 and 5
- 3 and 5
- primes

$-\frac{7}{3}$ lies between:

- -2 and -1
- -3 and -2
- 2 and 3
- -1 and 0

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. 7 as a fraction

ANSWER $\frac{7}{1}$

2. -4 as a fraction

ANSWER $-\frac{4}{1}$

3. 0.75 as a fraction

ANSWER $\frac{3}{4}$

4. $2\frac{1}{3}$ as a fraction

ANSWER $\frac{7}{3}$

5. $\frac{1}{4}$ as a decimal

ANSWER 0.25

6. $\frac{1}{3}$ as a decimal

ANSWER $0.333\ldots$

7. Is $\frac{5}{0}$ a number?

ANSWER No

Medium · the standard question

7 PROBLEMS

1. -1.6 as a fraction

ANSWER $-\frac{8}{5}$

2. $\frac{3}{8}$ as a decimal

ANSWER 0.375

3. $\frac{5}{6}$ as a decimal

ANSWER $0.8333\ldots$

4. $\frac{2}{11}$ as a decimal

ANSWER $0.1818\ldots$

5. Does $\frac{7}{20}$ terminate?

ANSWER Yes

6. Does $\frac{7}{30}$ terminate?

ANSWER No

7. $-\frac{7}{3}$ lies between

ANSWER -3 and -2

Hard · stretch and reason

7 PROBLEMS

1. Simplify $\frac{84}{126}$

   ANSWER $\frac{2}{3}$

2. 0.375 as a fraction in lowest terms

   ANSWER $\frac{3}{8}$

3. A number between $\frac{1}{3}$ and $\frac{1}{2}$

   ANSWER $\frac{5}{12}$

4. Order $\frac{2}{3}, \frac{3}{5}, \frac{5}{8}$

   ANSWER $\frac{3}{5} < \frac{5}{8} < \frac{2}{3}$

5. Which of $\frac{1}{6}, \frac{3}{16}, \frac{9}{25}$ terminate?

   ANSWER $\frac{3}{16}$ and $\frac{9}{25}$

6. Is every integer rational?

   ANSWER Yes

7. Is every rational an integer?

   ANSWER No

07 Homework

Homework — 5 tasks

Give every fraction in its simplest form.

1. Write 9, -2.4 and $3\frac{2}{5}$ as fractions.
2. Convert $\frac{5}{8}$, $\frac{4}{9}$ and $\frac{7}{20}$ to decimals.
3. Say which of them terminate, and why.
4. Simplify $\frac{72}{108}$.
5. Find a rational number between $\frac{2}{5}$ and $\frac{1}{2}$.